

Data Collected

+ EI_Analysis

+ participants-sheet

Study Recap

A 3-condition within-subjects study of a Chrome extension layered on Semantic Reader, testing whether elaborative interrogation prompts during academic reading improve comprehension.

- **Conditions:** Baseline PDF (A), Frictionless augmented (B), Productive friction with EI (C)
- **Design:** Latin-square within-subjects; each student experienced all three conditions across three weeks, one paper per week
- **Setting:** INFO 300 lab activity, two sections (BA, BB)
- **Group structure:** Lab groups of 3 students; conditions rotated weekly within group (2 groups with 4 students, with 2 students assigned the same condition each time)
- **Measures:** Per paper, an immediate quiz (5 items: 2 surface, 2 textbase, 1 situation model) and 7-day delayed quiz, plus an end-of-study survey covering preference, perceived learning, and perceived effort

3 types of questions in each quiz:

- Surface questions ask "what did the text say?"
- Textbase questions ask "what did the text mean?"
- Situation-model questions ask "what is the integrated mental model the reader built, and can they apply it?"

Summary: Data Collection

Overall

Metric	N
Enrolled	53
Consented (ideal)	35 (66%)
Non-consented who still completed at least one quiz	17

Complete cases (consented + all 6 quizzes)	25
End-of-study survey completion (among consented)	30/35 (85%)

By Section

Section	Consented / Enrolled	Consent Rate	Complete Cases
BA	18 / 28	64%	12
BB	17 / 25	68%	13

By Group

Group	Consent	Status
BA-G01	3/3	All consent
BA-G02	3/3	All consent
BA-G03	3/3	All consent
BA-G04	0/3	Zero consent; exclude entirely
BA-G05	3/3	All consent
BA-G06	1/3	Partial
BA-G07	2/3	Partial
BA-G08	2/4	Partial
BA-G09	1/3	Partial
BB-G01	2/3	Partial
BB-G02	3/3	All consent
BB-G03	2/3	Partial
BB-G04	0/4	Zero consent; exclude entirely
BB-G05	3/3	All consent
BB-G06	2/3	Partial

BB-G07	2/3	Partial
BB-G12	3/3	All consent

8 groups have full consent across all members. Two groups have zero consent and contribute no data.

Completion Among Consented (N=35)

Quizzes completed (of 6)	Students
6/6	25
5/6	3
4/6	7

Quiz	N (of 35)	Coverage
Paper 1 immediate	34	97%
Paper 1 delayed	32	91%
Paper 2 immediate	34	97%
Paper 2 delayed	33	94%
Paper 3 immediate	35	100%
Paper 3 delayed	25	71%

Within-subjects subset	N
All 3 immediate quizzes complete	33 / 35
All 3 delayed quizzes complete	25 / 35
All 6 quizzes complete (full within-subjects)	25 / 35

The Paper 3 delayed is the main coverage gap. Immediate-test analyses have 33-35 students; delayed analyses on Paper 3 drop to 25.

Inclusion Criteria

- **Consent:** Required
- **Completion:** At least 4 of 6 quizzes
- **End-of-study survey:** Not required for inclusion, but I have asked 5 people to complete
- **Resulting N:** 35 (every consented student meets the 4-of-6 threshold)

Final Dataset (210 rows): pid, group, week, paper, condition, timepoint, surface, textbase, situation_model, total_quiz, global_jol, intro_conf, gaps_conf, methodology_conf, findings_conf, discussion_conf, mental_demand, physical_demand, temporal_demand, effort, frustration, ei_helpfulness

A mixed-effects model handles their observed cells under maximum likelihood without bias if missingness is conditionally ignorable.

Two zero-consent groups (BA-G04, BB-G04) are excluded. These were also the lowest-engagement groups in the cohort.

Analysis

Analysis Plan

Data Preparation (ongoing)

- **A participant master sheet.** One row per consented student with participant ID, first name, section, group, and assigned condition per week.  participants-sheet
- **Score the 5 comprehension items per quiz form.** Surface (0-2), textbase (0-2), situation-model (0-1)
(0-3 via rubric)
 - Still need to think of a rubric for situation-model quiz responses.
- **Extract metacognition and load items per quiz.** Global JOL (0-100), 5 section-confidence ratings (7-point), 6 NASA-TLX-style ratings (7-point, immediate only), C-condition helpfulness (7-point, immediate and delayed separately).
- **Paper-ID.** A lookup table mapping each paperId to Paper 1, Paper 2, or Paper 3, for each group.

//////

- **Time-on-task.** Duration ranging from 431 seconds (7 min) to 3033 seconds (50 min). Decide on a minimum threshold for "actually read the paper", I suggest 600 seconds (10 min) as a lower bound, sessions below that get flagged for the data-quality filter.
- **Paste detection on C responses.** Flag and tag responses before EI quality coding.

Analysis

Outcome

Situation-model comprehension score (the construct H1 targets).

H1: Situation-model scores are higher in C than in A or B, on the immediate test. Surface and textbase show no condition difference or a small B advantage.

Model

Linear mixed-effects model:

score ~ condition + (1 | participant) + (1 | paper)

- **Fixed effect:** Condition (3 levels: A, B, C), treatment-coded with A as reference
- **Random intercepts:**
 - Participant: within-subjects variance
 - Paper: paper-difficulty variance (counterbalancing handles paper-condition at the design level)
 - Group: shared lab-discussion variance [NOT SURE ?] + (1 | group)

If the random-effects structure does not converge, drop in this order: group → paper. Participant intercept should stay.

Reporting

- Fixed-effect estimates with 95% CIs
 - Condition C produces situation-model scores 0.4 points higher than A on average, 95% CI
- Random-effect variances (especially group variance: low value = limited contamination)
 - Four numbers: how much score variance is between participants, between papers, between groups, and residual
 -
- Effect sizes (standardized condition contrasts)

≡ MixedModel

Secondary Analysis

- **Surface and textbase scores:** Same model, separate outcomes
- **Retention (immediate vs delayed):** Add time as within-subject factor and condition × time interaction.

H4 prediction: condition advantage for C grows with delay. Test on Papers 1 and 2 only, Paper 3 delayed N=25.

- **Metacognitive calibration (H3, partially testable)**
- **Cognitive load (NASA-TLX subscales, immediate only)**
 - 6 subscales: mental demand, physical demand, temporal demand, effort, frustration, performance
- **Perceived helpfulness of EI prompts (condition C only):**
 - Immediate vs delayed helpfulness ratings per student (Q9 immediate vs Q8 delayed, same C-condition session). Paired comparison: did helpfulness shift after a week?

- Cross-reference with performance: do students who rated prompts as more helpful actually score better on situation-model items in their C condition relative to their own A and B scores?
- **End-of-study survey (N=35):**
 - **Per-condition Likert (5-point) on four items per interface (A, B, C):** helped me understand, felt easy to use, retained more because of this, would choose again.
 - Friedman test for within-subjects comparison across conditions, per item. Follow up with Wilcoxon signed-rank pairwise tests, Bonferroni-corrected.
 - **Three forced-choice items:** which interface helped learn most, which for an exam, which felt easiest.
 - Test each against equal-preference null (chi-square goodness-of-fit, $df=2$).
 - **Direct H2 test (preference-performance dissociation):**
 - Cross-tabulate "helped learn most" against "felt easiest." A dissociation pattern means a sizable share of participants name different interfaces for the two. The H2 prediction: C (productive friction) wins "helped learn most" while B (frictionless) wins "felt easiest."
 - **Performance-rating triangulation:**
 - For each participant, compute their best-performing condition on situation-model items, then compare to their self-reported "helped learn most" choice. Disagreement is the strongest form of H2: objective performance does not match subjective belief about learning.
 - **Open-ended responses:** Four prompts (learning attribution, easy-vs-learning reasoning, effort comparison, general reflection).
 - Thematic analysis with two coders; report Cohen's kappa; theme frequency by condition.
- **EI engagement (extension logs, C condition):**
 - **Per-claim engagement status:** classify each of the 5 claims per session as engaged (panel opened + non-trivial response written), skipped (panel opened, no response), or unseen (panel never opened). Report distribution per participant and aggregate.
 - **Engagement metrics:** open count, dwell time per claim, first-keypress latency, total writing duration, response length (chars and words), reveal-click rate.
 - **Elaboration quality coding:** two raters independently code **each engaged response on the 1-3 scale (1 minimal/irrelevant, 2 partial, 3 integrative)**.

Report Cohen's kappa; resolve disagreements by consensus. Optionally cross-check with Claude-as-judge on the same rubric.

- **Paste detection (critical):** parse actionsLog per interaction. Flag any typing action where charCount jumps by more than ~30 chars in under 500 ms as a paste. Tag responses as **typed, partial_paste, or full_paste**. Report paste rate per participant and per condition exposure. Analyze elaboration quality separately for typed vs pasted responses; pasted responses are not evidence of genuine EI engagement.
- **AI-use sensitivity check:** if paste rates are non-trivial in C, run the primary comprehension model with paste rate as a participant-level covariate, and re-run on the subset of participants with zero pastes.

Sensitivity Analysis

- **Fully-consented-groups subset:** Refit the primary model on the 21 students in groups with full consent (BA-G01/02/03/05, BB-G02/05/12).
- **Complete cases only:** Refit on n=25 to check that the partial completers are not driving estimates.

Next Steps

- Develop a rubric to score situation-model responses
- Score both quiz forms for all three papers (rubric for situation-model items)
- Build the analytic dataset (long format: student × week × condition × paper × score + **engagement metrics**)
- Fit primary mixed model and run sensitivity analyses
- Two-rater coding of elaboration quality on C-condition typed responses
- Survey thematic analysis

Contributions

A deployment study with a within-subjects design

1. **Empirical (constrained):** First in-the-wild within-subjects test of EI-augmented academic paper reading.
2. **Methodological:** Patterns from authentic classroom deployment of a productive-friction intervention: consent dynamics, completion attrition, group-level contamination, what is preserved and what is lost when moving from lab to course context.
3. **Design artifact:** The Chrome extension itself as a design contribution. Architecture, claim extraction, EI prompt generation, expert-answer presentation.
4. **Process insight:** How students actually engage with EI prompts during real academic reading, including the gap between prompt presentation and meaningful elaboration.

<https://www.solaresearch.org/events/lak/>

<https://learningatscale.hosting.acm.org/las2026/>

Open Questions

1. Inclusion threshold: 4-of-6 quizzes feels right given that it doesn't exclude anyone. Worth discussing whether to also report a complete-cases-only result as the primary?
2. Venue target: which framing leads the writeup (deployment story, design artifact, empirical effect)?
3. Should we include the EI engagement analysis?
4. Paste detection on C responses: sample shows one student pasted long responses on 2 of 5 claims. How many pasted responses across the cohort, and how do we handle them in the elaboration quality analysis? Drop, flag, or analyze separately?

High-level observations

High-level observations

- Students pasted long-form responses into the EI prompts (jumped to 1028 chars within 199 ms)
- Some prompts, they skipped answering
- If students pasted EI responses, some likely also used AI. same for the quiz items. So, the EI response quality may not be actually judging the quality.
- can report paste rates per participant – jumps by more than roughly 30 characters in under 500 milliseconds is a paste
- Time on task is wildly variable.

Tab 4

Formal hypotheses

H1 The productive friction (EI) condition will produce higher situation-model comprehension scores than both the frictionless augmented and baseline PDF conditions on the immediate test.

Surface and textbase scores will show no significant condition differences or small frictionless-augmented advantages

H2 (Preference-performance dissociation). Participants will rank the frictionless augmented condition as most preferred and the productive friction condition as least preferred (or second-least)

H3 (Metacognitive calibration). Goodman-Kruskal Gamma correlations between item-level JOLs and comprehension performance will be significantly higher (better calibration) in the productive friction condition than in either comparison condition. Additionally, calibration bias (mean JOL/mean performance) will be smallest in the productive friction condition, and participants will show less overconfidence.

H4 (7-day retention differential). The advantage of the productive friction condition over the other two conditions will be **larger** at the 7-day delay than at the immediate test, specifically for situation-model questions. Surface-form scores will converge toward chance across all conditions at delay (consistent with Fisher & Radvansky, 2018: surface A' near chance by 1 hour). Textbase scores will show moderate decay across all conditions. Situation-model scores in the EI condition will show the least decay.

H5 (Prior knowledge moderation). The benefit of EI on situation-model comprehension will be moderated by prior knowledge: participants with higher prior knowledge scores (Phase 1) will show a larger EI advantage than those with low prior knowledge.

H6 (Elaboration quality mediation). Participants who produce higher-quality elaborations (rated 3 on the 1-3 scale) will show larger situation-model gains than those producing low-quality elaborations (rated 1), controlling for prior knowledge.

Tab 5

